

AI Audit Report

Assignment: HW05 - AI-assisted Performance Testing Student ID: 23127184 System under test: EShop REST backend AI tool: OpenAI Codex Audit period: 2026-08-16 to 2026-08-18 (Asia/Saigon)

1. Declaration

I used OpenAI Codex as an engineering assistant to review the assignment, improve reusable Agent Skills, design JMeter scenarios, automate repeatable test execution, analyse raw JTL data, challenge initial interpretations, and prepare reviewable documentation. I remained responsible for the final test design, criteria, interpretation and submission.

The performance measurements in this assignment were not invented by AI. They were produced by the local EShop backend, Apache JMeter and the Windows resource monitor on host KIEUDUYEN. Raw JTL files, resource samples, screenshots, JMeter logs and HTML reports are retained in the repository so that every material claim can be checked independently.

2. Prompt and evidence policy

The prompts below document the task instructions used in the working sessions. Where one activity required several follow-up messages, the instructions are consolidated into one interaction record so the technical objective, constraints and expected deliverables remain readable. The Codex conversation history is the supporting source for the complete message sequence.

AI output is summarized rather than copied as an entire transcript. Each summary is tied to a file, raw result directory or Git commit. Where AI made an incorrect or overconfident claim, the correction is recorded explicitly instead of silently replacing the original interpretation.

3. Interaction register

ID	Date	Purpose	Main verified outcome
-:	-----	-----	-----
01	2026-08-16	Requirements and Agent Skill review	Evidence, portability and analysis gaps identified
02	2026-08-16	Agent Skill hardening	Reusable five-skill workflow validated and committed
03	2026-08-17	Environment, baseline and criteria	Real host baseline and pre-registered thresholds recorded
04	2026-08-17	Scenario planning	Measured Load, Stress, Spike and Soak profiles designed
05	2026-08-17	JMX implementation and pilot	Four portable plans generated; partial-journey bug found
06	2026-08-17	Official execution	Four complete genuine result sets retained
07	2026-08-17 to 2026-08-18	Raw analysis and critique	AI misinterpretations corrected from JTL and source code
08	2026-08-18	Continuous testing and reporting	Main report, critique, flow chart and PDFs produced
09	2026-08-18	Submission audit and AI Audit rewrite	Public branch and machine-checkable deliverables verified

4. Detailed interaction log

Interaction 01 - Review the requirements and existing Agent Skills

Field	Record
Tool	OpenAI Codex
Date	2026-08-16 (Asia/Saigon)
Objective	Establish a rubric-traceable workflow before changing or running anything
Primary inputs	refs/requirements/ver2.md, course guidance, .claude/skills/

Prompt

Read the complete HW05 specification and inspect every performance-testing Agent Skill, reference, script and asset already present in the repository. Map the seven taught performance-testing steps and every rubric requirement to concrete artifacts. Identify missing evidence gates, unsafe assumptions, portability problems and analysis errors. Do not execute an official test or fabricate any screenshot, log, timestamp or measured value. Return a prioritized correction list with the files that should change.

Condensed AI output and actions

- Mapped environment, criteria, planning, implementation, execution and analysis responsibilities across the reusable skills.
- Identified an unsuitable linear Stress shape, machine-specific CSV paths, ambiguous Node-process selection and incomplete resource/evidence gates.
- Found that the analyzer could mishandle final partial time buckets and could trust Transaction Controller rows too readily.
- Recommended a dedicated HW05 evidence contract to separate genuine execution evidence from AI-generated documentation.

Human verification and decision

The findings were accepted only where they matched the official brief and the repository. No official result was claimed at this stage. The review became the input for Interaction 02.

Trace: .claude/skills/performance-testing/, commit e385ad7.

Interaction 02 - Harden and validate the reusable Agent Skills

Field	Record
Tool	OpenAI Codex
Date	2026-08-16 (Asia/Saigon)
Objective	Make the skills safe, reusable and capable of producing auditable HW05 artifacts
Primary inputs	Findings from Interaction 01 and the existing skill scripts/assets

Prompt

Implement the approved corrections in the performance-testing Agent Skills. Use a four-stage accumulating Stress profile, a portable runtime data.dir, live CSV validation, explicit backend-PID monitoring and safe monitor cleanup. Correct tail-bucket throughput calculations and make memory warnings cautious rather than calling growth a leak. Add an HW05 evidence contract that forbids fabricated evidence. Validate all skill files and scripts, perform only a technical smoke check where appropriate, and commit the scoped changes with a Conventional Commit message and no co-author trailer.

Condensed AI output and actions

- Implemented the four-stage Stress design and portable JMeter data path.
- Added CSV preflight checks and mandatory explicit backend PID selection.
- Hardened resource-monitor cleanup and final-bucket duration handling.
- Added cautious memory interpretation and the HW05 evidence contract.
- Validated the five skills and supporting scripts, then created commit

e385ad7 feat(hw05): harden performance testing skills.

Human verification and decision

The student reviewed the commit scope and required a normal Conventional Commit without Co-authored-by. Unrelated worktree content was excluded.

Trace: .claude/skills/, evidence/git-commit-log.txt.

Interaction 03 - Establish the environment, baseline and criteria

Field	Record
Tool	OpenAI Codex
Date	2026-08-17 (Asia/Saigon)
Objective	Measure the actual local system before selecting load levels
Primary inputs	Student ID 23127184; confirmed unique purchase workflow; local EShop SUT

Prompt

Begin HW05 for student 23127184 using the confirmed non-duplicated workflow: Login -> Search product -> Product detail -> Add to cart -> Checkout. Inventory the real host and tool versions, document the co-located load-generator limitation, start a fresh backend with LOADTEST=1, create deterministic test data through the live API, validate every CSV row, and measure a repeatable single-user baseline. Propose label-specific p95, error and journey-completion criteria before any official run. Clearly separate measured facts from sizing assumptions and preserve the commands and evidence paths.

Condensed AI output and actions

- Recorded hostname KIEUDUYEN, CPU, RAM, Windows, Java, JMeter and Node.js

versions.

- Started a fresh local backend, created and validated 240 data rows in each CSV

pool and measured seven single-user samples per endpoint.

- Calculated a 30.4 ms workflow service baseline and documented the co-located

generator constraint.

- Pre-registered p95 limits of 250 ms for login, 200 ms for reads, 300 ms for

cart/checkout and 1,000 ms for the complete journey, with error rate below 1%.

Human verification and decision

The thresholds were reviewed before the official results were known. The student confirmed that the selected workflow did not duplicate another group member's work. Hardware screenshots and same-frame execution evidence remained mandatory human-attribution items.

Trace: docs/phases/01_environment.md, docs/phases/02_criteria.md, evidence/baseline/, commit c18293b.

Interaction 04 - Plan realistic Load, Stress, Spike and Soak scenarios

Field	Record
Tool	OpenAI Codex
Date	2026-08-17 (Asia/Saigon)
Objective	Convert the measured baseline into defensible scenario profiles
Primary inputs	Baseline, fixed criteria, 240-row CSV pools, 8 GB co-located host

Prompt

Design Load, Stress, Spike and 10-15 minute Soak scenarios for the confirmed five-step workflow. Derive concurrency from the measured service time and a disclosed compressed think time instead of choosing round numbers. For each scenario, specify VU shape, ramp, stable duration, expected arrival rate, question answered, pass/fail criteria and required evidence. Assign three different listener types across Load, Stress and Spike. Explain data recycling, account-lockout handling, reset requirements and the limitations of running JMeter beside the backend. Include a separate human review of what the initial AI defaults got wrong and why.

Condensed AI output and actions

- Derived Load at 34 VU, Stress stages at 33/66/99/132 VU, Spike at 17 baseline plus 168 burst VU, and Soak at 27 VU.
- Used 300-700 ms uniform think time and explicitly described it as a compressed arrival-rate model rather than literal browsing behavior.
- Assigned Summary Report, Aggregate Report and error-only View Results Tree to the three required scenarios.
- Documented reset, resource, screenshot, raw-log and invalid-run gates.

Human verification and decision

The review rejected inherited 50/200/200/40-VU defaults, a linear Stress ramp, machine-specific paths, weak status-only assertions and process-name-only monitoring. The reviewed design was accepted for implementation.

Trace: docs/phases/03_plan.md, docs/reviews/01_ai_plan_review.md, commit 8087833.

Interaction 05 - Generate the JMX plans and review a diagnostic pilot

Field	Record
Tool	OpenAI Codex
Date	2026-08-17 (Asia/Saigon)
Objective	Verify correlation, assertions, data and evidence tooling before official execution
Primary inputs	Approved profiles, JMX assets, CSV data and runner scripts

Prompt

Generate four portable JMeter plans named with student ID 23127184, scenario type and date 20260817. Implement JWT extraction, dynamic product correlation, data-driven request bodies and content assertions for token, product ID and order ID. Validate the XML and CSV references. Then run a short 2-VU/30-second diagnostic pilot only, preserving its JTL, HTML report, JMeter log, run record and backend resource samples. Analyse the pilot from raw CSV-form JTL, compare controller rows with all five endpoint-label counts, document every correction and do not present pilot numbers as official results.

Condensed AI output and actions

- Generated Load, Stress, Spike and Soak JMX files with portable data paths and the approved workflow/assertions.

- Completed a diagnostic pilot with 117 request samples, 0% errors and 17 ms request p95.

- Initially observed 23 successful controller rows, then found only 22 journeys containing all five endpoint labels because the scheduler ended a tail iteration.

- Corrected the analyzer to report endpoint-complete journeys and expose tail/ambiguous controller rows.

Human verification and decision

The pilot was classified only as implementation validation. The human review rejected the misleading statement "23/23 transactions passed" and adopted the conservative 22 complete-journey count for all later analyses.

Trace: test-plans/, evidence/pilot-runs/, docs/reviews/02_pilot_review.md, commit 8087833.

Interaction 06 - Execute and preserve the official scenarios

Field	Record
Tool	OpenAI Codex
Date	2026-08-17 (Asia/Saigon)
Objective	Produce genuine, attributable and reproducible execution evidence
Primary inputs	Four reviewed JMX plans and the official execution checklist

Prompt

Execute the reviewed scenarios in order: Load, reset, Stress, reset, Spike, reset, then Soak. Before every run, start a fresh backend with LOADTEST=1, record its PID, regenerate and validate all 240-row CSV files, verify the API, confirm safe free memory and monitor only that backend PID. Preserve the full timestamped result.jt1, HTML report, resources.csv, jmeter.log and run.md. Capture genuine same-frame JMeter and Task Manager screenshots. Stop and retain a run as invalid if preflight, monitoring or evidence capture fails. Never edit raw results or infer evidence that was not captured.

Condensed AI output and actions

- Completed and retained four official result sets for Load, Stress, Spike and Soak.

- Preserved 42,810 / 94,104 / 28,325 / 49,727 request samples respectively, with 0% request/assertion errors.

- Captured real backend CPU/memory logs, hardware identity and same-frame screenshots.

- Committed official plans and evidence in 98462cf test(hw05): capture official performance results.

Human verification and decision

Raw evidence was retained unedited. The student remained responsible for the required Vietnamese narration and YouTube upload; AI did not generate or claim that evidence.

Trace: results/, evidence/screenshots/, evidence/reset-records/, commit 98462cf.

Interaction 07 - Analyse raw JTL data and hunt AI misinterpretations

Field	Record
Tool	OpenAI Codex
Date	2026-08-17 to 2026-08-18 (Asia/Saigon)
Objective	Recompute ground truth and challenge AI-generated conclusions
Primary inputs	Four raw JTL files, four resource logs, fixed criteria and relevant backend code

Prompt

Analyse every official raw JTL as proper CSV. Separate request samples from Transaction Controller rows; calculate count, throughput, mean, p50, p90, p95, p99, max, bandwidth, errors, per-label values, 30-second phases and endpoint-complete journeys. Identify the Stress knee only if throughput stops scaling while latency or errors rise. Compare Spike baseline, burst and recovery separately. Treat Soak memory as a trend, not automatic proof of a leak. Then review the AI analysis line by line, cite corrected raw values and classify each optimization as supported, plausible but unverified, or unfounded after checking the source code.

Condensed AI output and actions

- Recomputed overall request p95 values of 10/11/19/11 ms for Load/Stress/Spike/Soak and 0% errors in every run.
- Corrected controller overcounts: 8,557/18,797/5,622/9,939 controller rows versus 8,547/18,775/5,591/9,934 endpoint-complete journeys.
- Found no Stress knee through 132 VU; the highest stable stage reached about 260.6 request/s at 13 ms p95 with 0% errors.
- Measured Spike burst at 361.6 request/s and 26 ms p95, with recovery to 12 ms in the first full recovery bucket.
- Demonstrated a Soak threshold of at least 54.5 request/s at 27 VU with a 172.0 MB memory ceiling, while rejecting an unsupported leak conclusion.
- Classified cart clearing as supported, email indexing and SQLite WAL as plausible but unverified, and a generic connection pool plus a normal B-tree index for LIKE '%keyword%' as unfounded for this code path.

Human verification and decision

The reviewed report uses the conservative endpoint-complete journey count and does not claim that 132 VU is capacity or that memory growth proves a leak. Every material correction is linked to raw data or source behavior.

Trace: docs/analysis/, docs/reviews/03_ai_analysis_review.md, docs/phases/04_analysis.md, commit 7327273.

Interaction 08 - Propose continuous testing and build the reports

Field	Record
Tool	OpenAI Codex
Date	2026-08-18 (Asia/Saigon)
Objective	Convert the reviewed evidence into complete submission documents
Primary inputs	Reviewed analysis, official rubric, AI Critique requirement and result evidence

Prompt

Draft the final HW05 report from reviewed evidence only. Include environment, fixed criteria, AI-assisted design corrections, scenario results, endurance threshold, limitations, AI misinterpretation hunt and optimization judgement. Propose a commit-aware continuous-performance pipeline with risky-path filtering, a short Load gate, a confirmation run, label-specific p95 comparison, scheduled Stress/Spike and release-candidate Soak. Discuss cost, false positives and false negatives, and include a flow chart. Write a 200-300-word AI Critique, update the AI Audit, render the main report and Audit as polished PDFs, and visually inspect every page for clipping, blank pages and unreadable tables.

Condensed AI output and actions

- Produced the main Markdown report, continuous-performance proposal and flow chart, plus a 272-word AI Critique.

- Built an eight-page performance report PDF and the then-current three-page AI Audit PDF, which Interaction 09 later replaced with the expanded final Audit.

- Detected and fixed a blank trailing Audit page and a clipped Interaction heading during visual QA.

- Added a Vietnamese video guide and a fail-closed packaging script that refuses to create a final ZIP without a real YouTube URL.

Human verification and decision

The AI Critique is retained as an AI-assisted artifact, while the narrated demo is treated as separate student-authored evidence. The packaging guard rejects an absent video URL.

Trace: 23127184_HW05_REPORT.md, docs/AI_CRITIQUE.md, docs/phases/05_continuous_performance.md, output/pdf/, commits 3770903 and b9229a1.

Interaction 09 - Audit submission readiness and rewrite this AI Audit

Field	Record
Tool	OpenAI Codex
Date	2026-08-18 (Asia/Saigon)
Objective	Make the final audit readable, complete and defensible without altering measured evidence
Primary inputs	Official brief, all review files, Git history, final PDFs and public branch

Prompt

Audit the current HW05 repository against the official submission checklist. Verify JMX validity, distinct listeners, 240-row CSV inputs, raw JTL/resource counts, HTML report presence, PDF rendering, Git history and public branch visibility. Then replace the existing AI Audit completely with a professional report. Rewrite the working prompts so they are clear, specific and technically correct. Preserve the true chronology, AI mistakes, human corrections, artifact paths and commit traceability. Do not fabricate the missing video, student narration, issue report or any measured result.

Condensed AI output and actions

- Verified all four JMX files as valid XML and confirmed the required distinct listeners across Load, Stress and Spike.

- Confirmed three 240-row CSV pools, four raw JTL/HTML/resource result sets and recomputed 0% error with p95 values matching the reviewed report.

- Verified public branch hw5 and synchronized local/remote commit state.

- Replaced the former short-form Audit with this structured interaction log and

linked every material result to its evidence or commit.

Human verification and evidence boundary

The rewritten Audit preserves prompt intent, raw-result corrections and artifact traceability. The narrated video and any oral-defence explanation remain student-authored evidence and are not generated or inferred by AI.

Trace: `scripts/package-submission.ps1`, public branch `hw5`.

5. Known limitations

- This document consolidates multi-turn working sessions into traceable interaction records. The Codex conversation history contains the complete message sequence.
- The Load generator and backend shared one 8 GB laptop, so the demonstrated throughput is specific to this environment and is not production capacity.
- The rate limiter was deliberately bypassed with `LOADTEST=1`; the runs measure application performance under that disclosed configuration.
- No genuine performance failure occurred, so no GitHub performance issue was fabricated.
- Student narration and personal review are attribution evidence outside the scope of AI-generated artifacts.